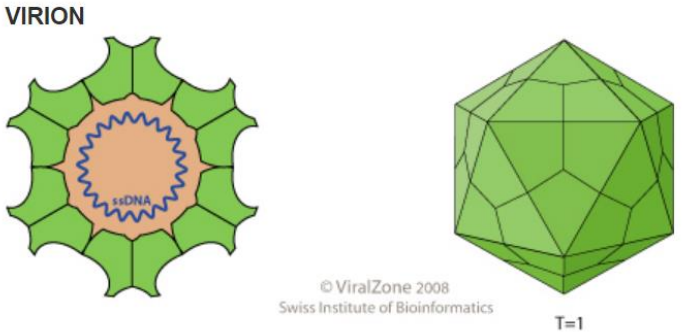
Chicken Infectious Anemia
雞傳染性貧血症

Nai-Huei Wu, DVM, PhD



introduction

- 1979 – Japan, by Yasua *et al.,*
- 曾被叫做chicken anemia agent (CAA), parvovirus-like virus (PVLV)
- 台灣於1991年首次分離到病毒而證實台灣雞場有此病
- 再生不良貧血、淋巴萎縮性、免疫抑制疾病
  - Anemia
  - Lymphoid depletion
  - Liver
  - Hemorrhages (皮下、肌肉)
  - In both **Layer and Broiler**



Cause

- Cause by **chicken infectious anemia virus** (CIAV, or CAV)
  - 台灣的共識

    病名: 雞傳染性貧血 (CIA)

    病原: 雞貧血病毒 (CAV)
- Commonly found in poultry farm – **worldwide distribution**
- All ages of chicken can be infected
  - **<2 weeks old chick → severe clinical signs**
  - 一日齡小雞感受性最強
- 雞是唯一宿主，Turkeys may infect as well

Importance

- Trans-ovarian transmission
- Immunosuppression
- 傳播效率 – 中 (需數週方可傳播至全場)
- 潛伏期 – 長 (需數週才可發現臨床症狀)
- 增加其他疾病嚴重程度
- 影響育成率、換肉率、產蛋率

Economic impact – Broiler

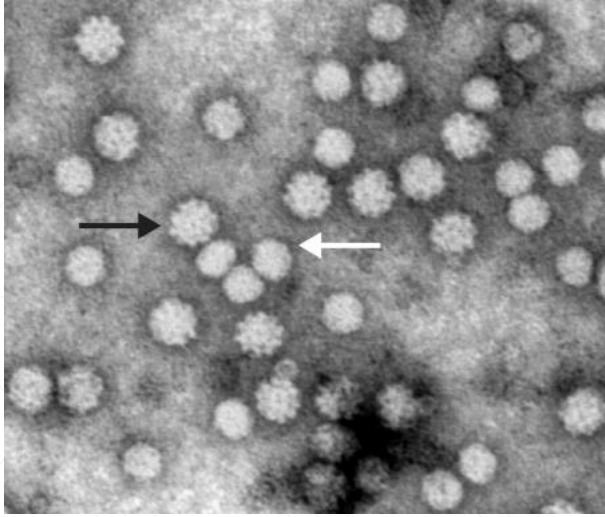
- Bad performance *飼料換肉率低下
  - Subclinical CAV infection → flocks appear normal (但吃不胖)

- Condemnation rate increase (屠宰時的廢棄率上升)
- Secondary infections
- Vertical transmission

Economic impact – Broiler

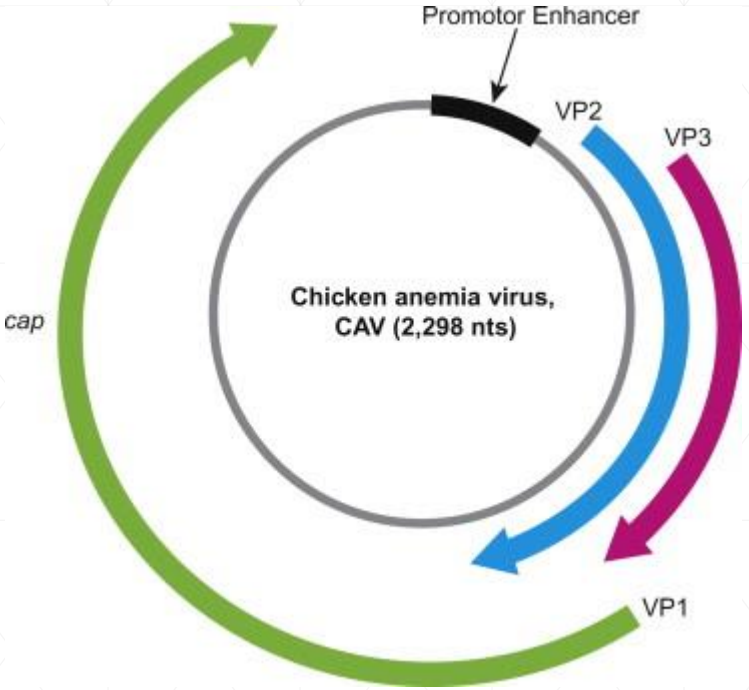
CAV status of farm	net income	Weight	Mortality
Clean	+2.4%	+0.8%	-0.16%
Infected	**-10.6%**	**-1.7%**	**+0.24%**

Etiology



- Circovirus
- **Smallest virus**
- Non-enveloped, icosahedral (20面體), 25nm
- Family: *Anelloviridae* (指環)
- Sub-family: *Circoviridae* (環狀病毒)
- Species: Chicken anemia virus (CAV)
- Other circovirus infection in avian
  - 鸚鵡: PBFDV
  - 鴨、鵝: DuCV、GoCV

Virus structure



- **Circular**, **ssDNA (-)**
- Genome: 2.3kbp (Avipox=300kbp)
- Two types: **CIAV-I**, CIAV-II.
- One serotype

- 3 open reading frames:
  - VP1 **Capsid** (structure protein)
  - VP2 (non-structure, multifunction)
  - VP3 **apoptin** – induct apoptosis (been use to anti-cancer)

Virus facts

- **Stable 對環境抵抗性強**
- does not inactivate :
  - pH 3
  - Chloroform (氯仿)
  - 70度C x 1小時 或 80度C x 5分鐘
- Resistant to Lipid solvents
- Resistant to common Disinfectants
- 碘劑,次氯酸, 37℃, 10%高濃度, 2h

Transmission



年輕母雞

Vertical (Hens ♀)



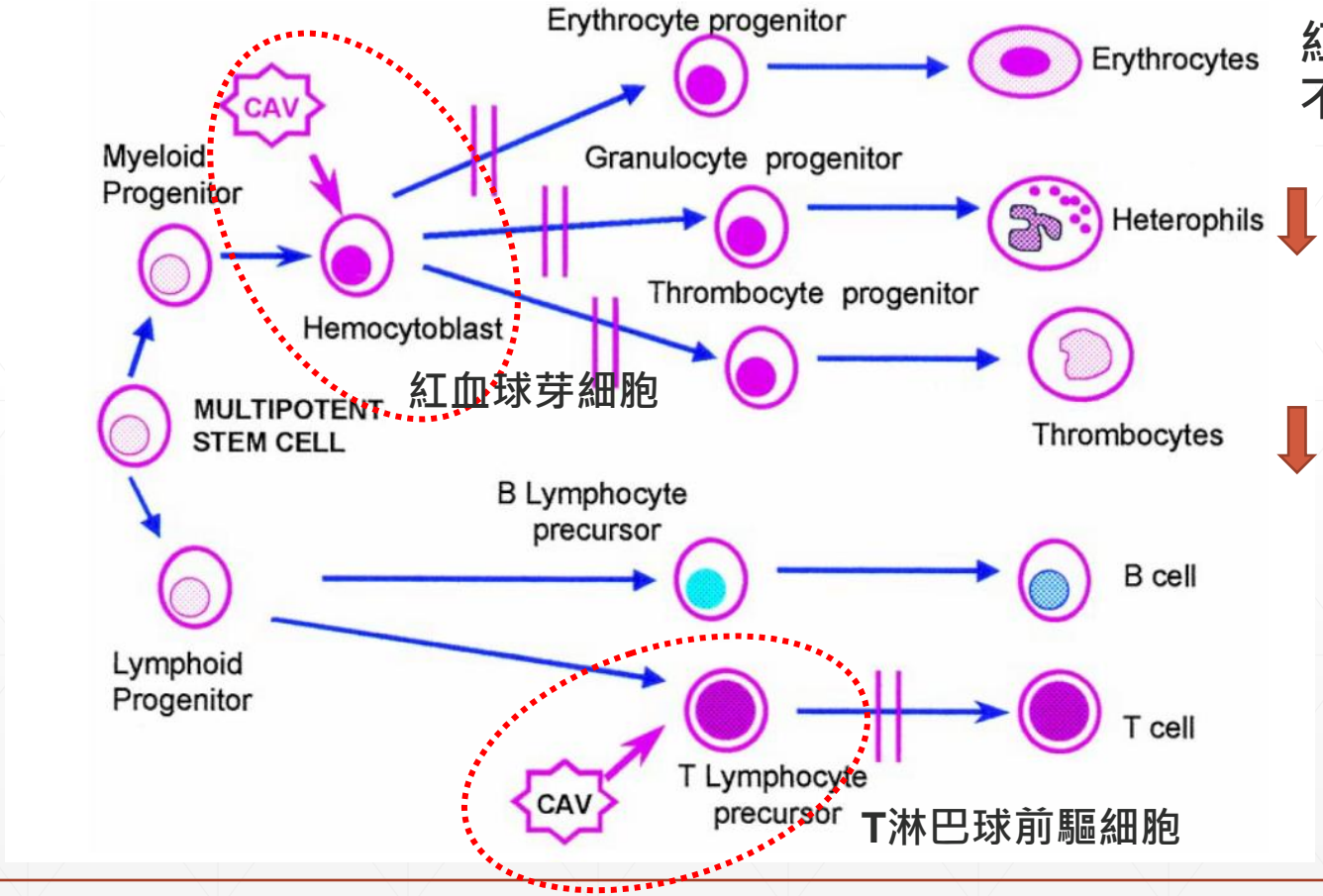
Horizontal

糞口/機械



- 疾病發生的時期非常早 – 小雞孵化後三週內
- **Vertical transmission**
  - Young flocks → Egg
  - Infected during the production period
- Horizontal transmission
  - Organic material
  - Contaminated equipment
  - 呼吸道、消化道、皮膚創傷、人工授精亦可成立

Effects of CAV on hemopoiesis and T cell development



Organ affinity

- Virus disseminated all over the body.
- Isolation of virus from different organs – Possible
- **Thymus** cannot used to isolate virus – target organ.
  - Due to the impairing the T cell maturation.

Age resistance

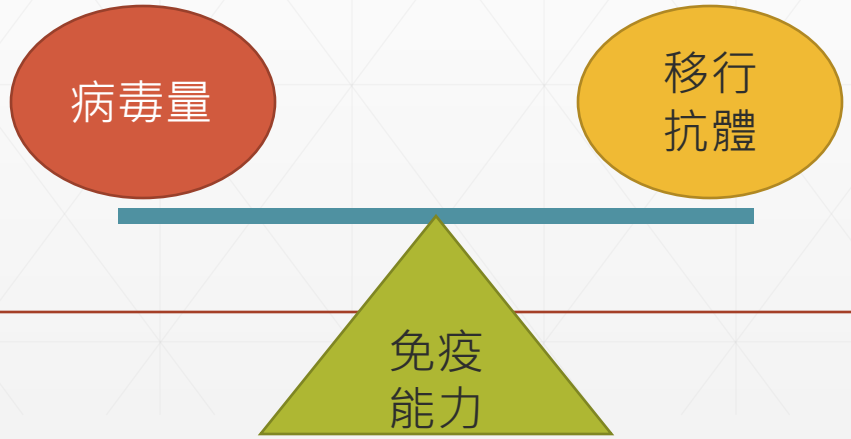
- Age < 2weeks – High morbidity / High mortality
- Age > 3-6weeks 開始– "**Age resistance**"
- **1日齡小雞感受性最強**

- Age resistance, but **still** "could be infected"
- Age resistance related to other immunosuppress pathogens
  - IBD, MD, Reo (selected) →可能增加 CAV致病力→延遲age resistance

Younger birds present clinical signs

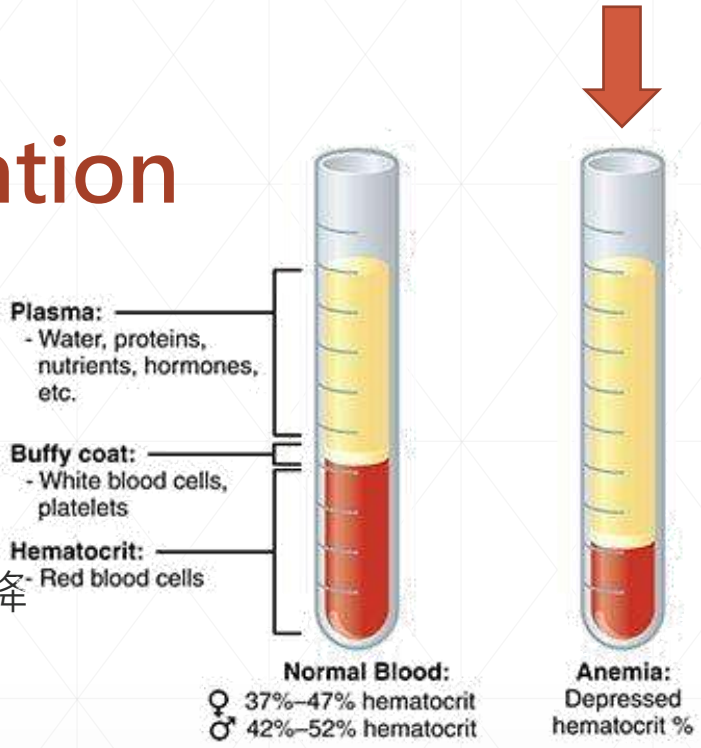
小雞是否感染 跟

- 1. 病毒量
- 2. 移行抗體
- 3. 是否有其他免疫抑制性疾病 (IBD, MD, REO) 有關
- 4. 好發於 更換**新母雞**後 的 小白肉雞



Post infection complication

- After infection 8 days post infection
  - **Hematocrit levels** 血容比 下降
  - **Thrombocytes** 血小板數 下降
  - **Red, White cell counts** 白/紅血球數下降



- 28-36 days post infection
  - Back to normal

臨床症狀

- Anorexic 厭食, Depressed 消沉
- **Huddle** (縮在一團) under the heat source. 畏寒
- 發育不良, 飼料換肉率差
- Exhibit a marked **pallor 蒼白**
  - **貧血→臉部 雞冠 肉垂蒼白**
- Secondary infections

臨床症狀

- Hemorrhages 易出血 – muscle, subcutaneously, “Wing tips”
  - (血小板減少)
- Bone marrow – pale or yellow, Fatty consistency.
  - (代償性)

- **Thymic atrophy**
- **Bursal atrophy** (in 10-14 day-old, **vertically** infected chickens)

外觀





感染 CAV 之 8 週齡蛋中雞，呈現沉鬱、厭食、臉部及雞冠蒼白等臨床症狀。

Blue wing disease







Subcutaneous hemorrhages in the wing and hock joint.
→ Blue discoloration.
→Blue wing disease

解剖病變

- Ecchymosis (瘀青) skin hemorrhages (mostly wings)
- Lesions – turn blue and break

- 5-6 days after clinical signs → mortality peaks


c

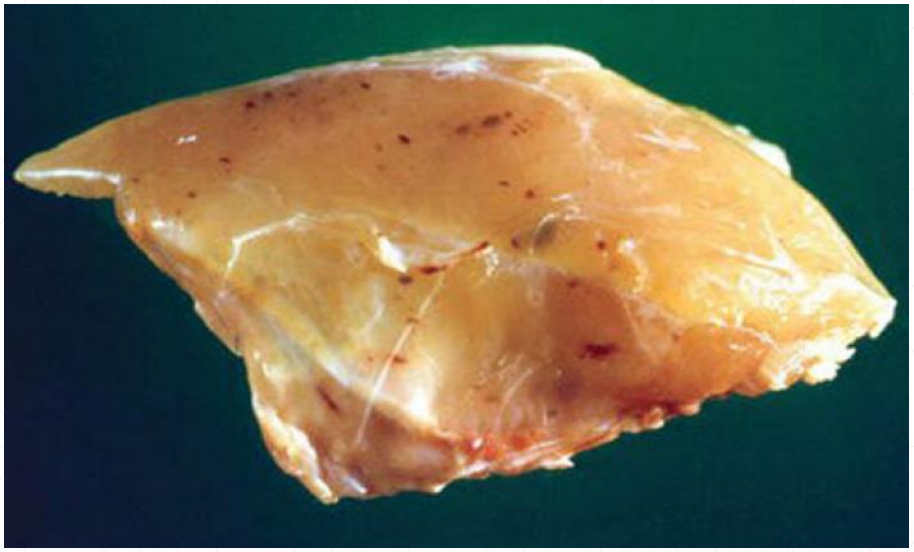

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Muscles





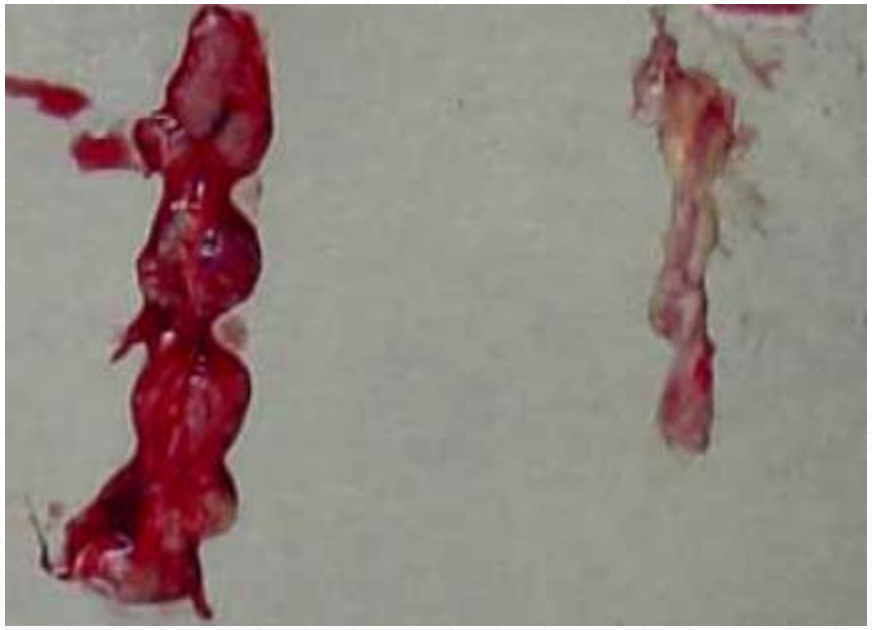
- Hemorrhages in breast, thigh and leg muscles
- 有可能是其他疾病



Immunosystem



- 脾臟轉為淡紅色



胸腺淋巴萎縮



Bone marrow

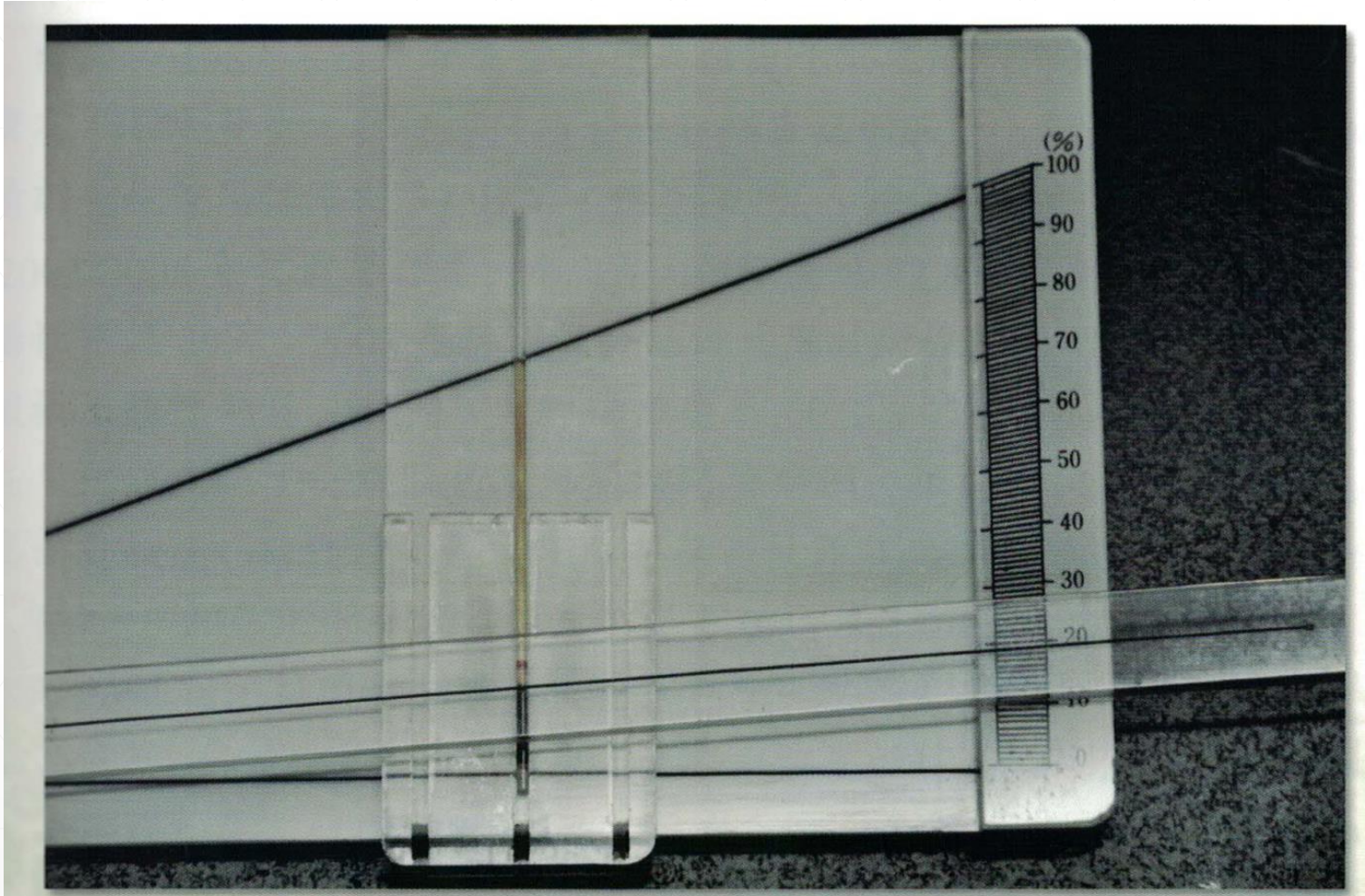




- Femur with normal dark red bone marrow (top)
- femur with **pale aplastic bone marrow** (bottom)



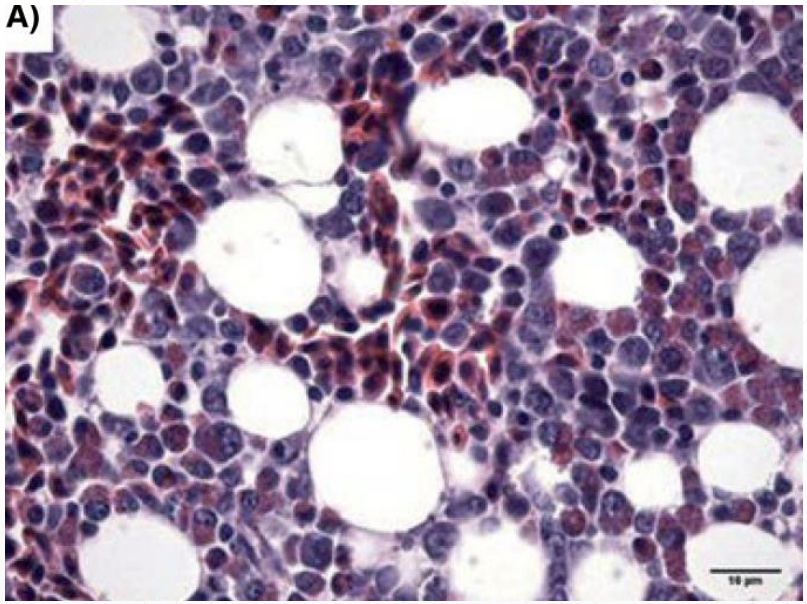
圖五　骨髓細胞被脂肪細胞所取代，呈黃骨髓樣病變。



圖六　血溶比僅 19%，呈現低紅血球數現象 (正常應大於 27%)。

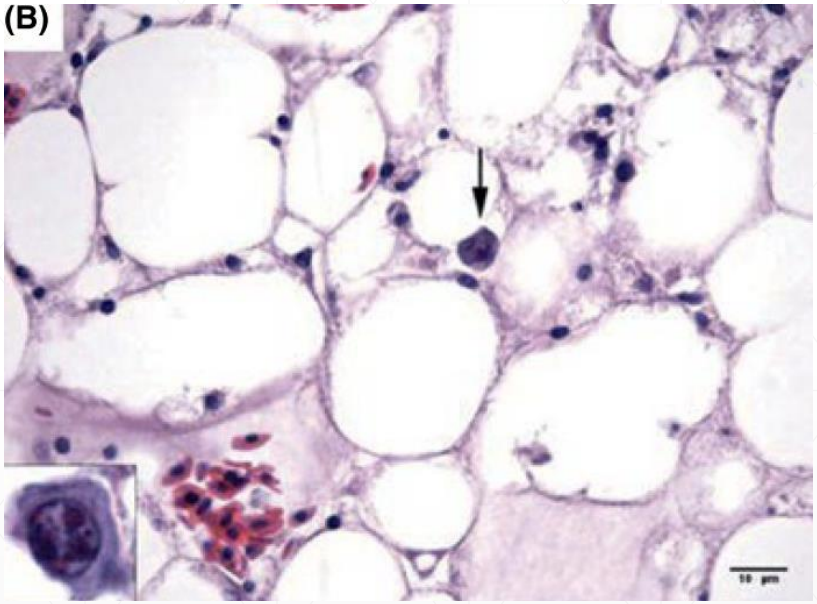
Histopathology

Normal



(A) Normal femoral bone marrow

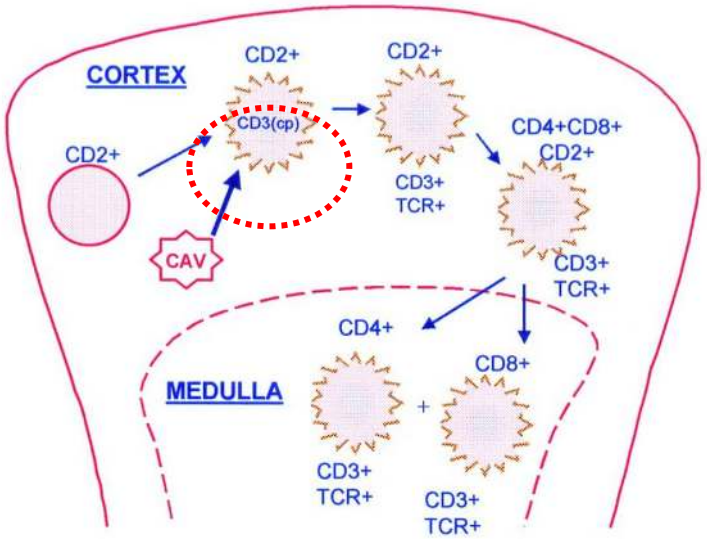
CIA

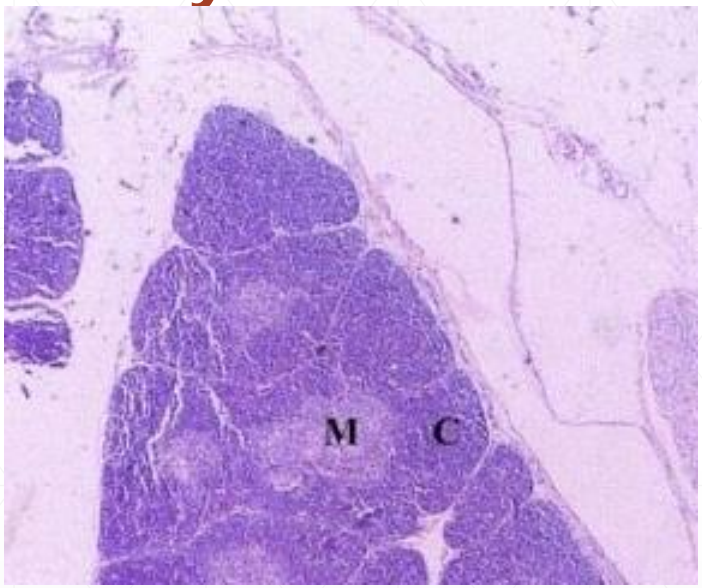


(B) Femoral bone marrow with CIAV

Note atrophy of hematopoietic and myeloid tissue and presence of fat cells. Few cells have small intranuclear inclusion bodies (arrow)

Immunosuppress – in Thymus

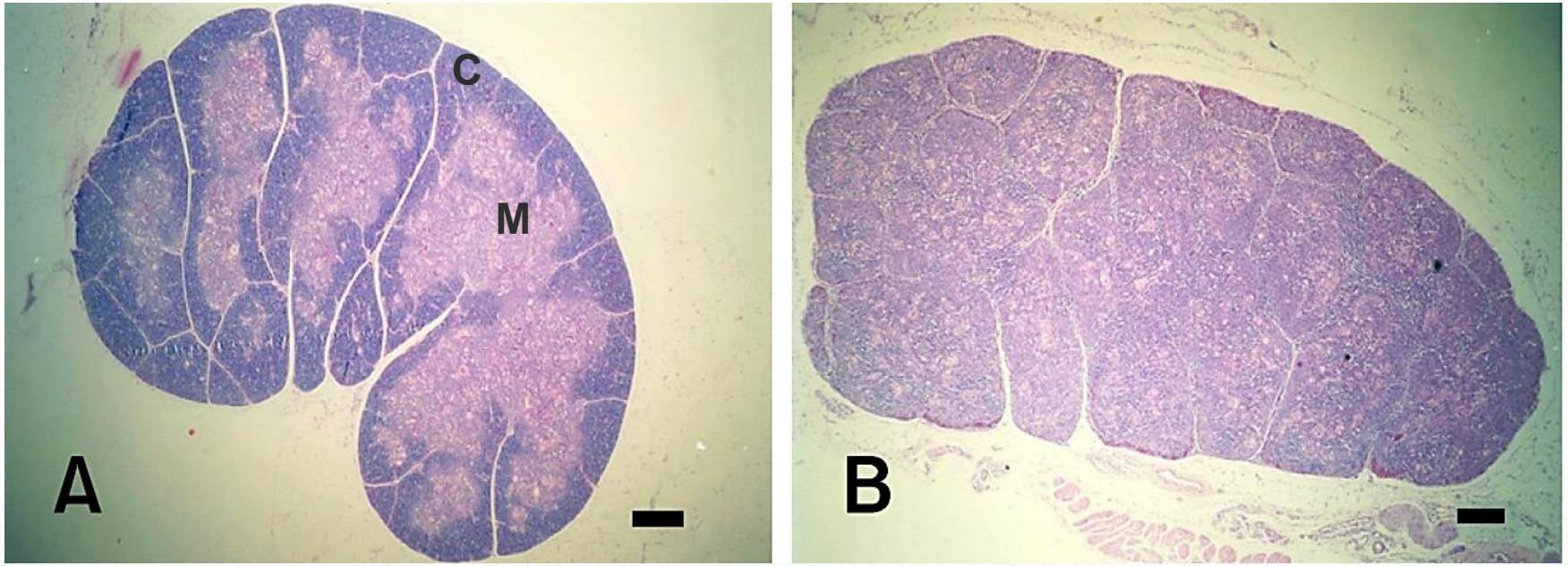




Targeting of T cell precursor cells in the thymus.

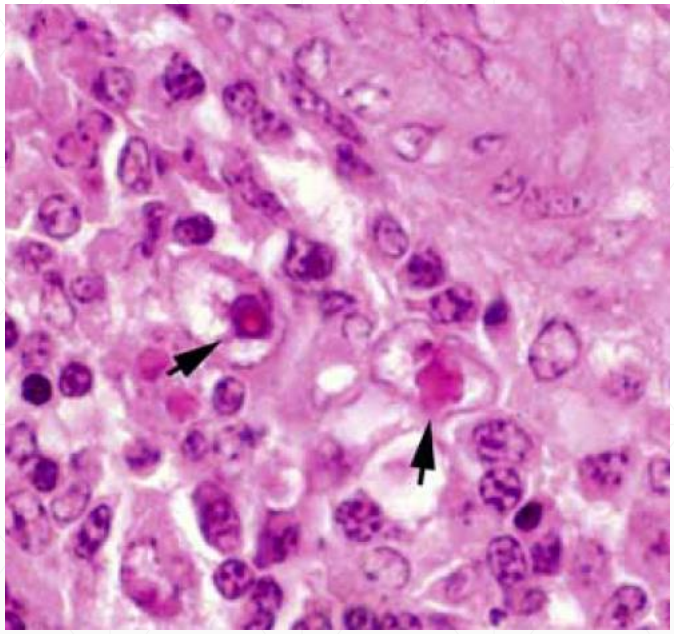
- Major depletion in the thymic cortex
- Due to destruction of precursor cells.
- Cells expressing *cytoplasmic* CD3 are the earliest cells infected.
- CD8 Cytotoxic T cell immature

Thymic atrophic



- Depletion of the cortical thymocytes and medulla-like appearance of the thymic cortex

Spleen



Splenic intranuclear inclusions.

Eosinophilic, intranuclear in lymphocytes

Diagnosis

- 血容比　　PCV ≤27
- 雞隻外觀 – 蒼白 (<3week old雞)
- 區別診斷 – MD, REV, IBD



CIA

- 實驗室檢查
  - PCR
  - IHC
  - 病毒分離 - MSB-1細胞 (chicken lymphoblast)
- 血清學診斷代表意義不足 (immunosuppress)

Vaccine

- 減毒疫苗
- 打母雞 (種雞) - 利用移行抗體保護小雞
- 8週齡-開產前4W，初產 或是 "更新"母雞
- 不可施打於 小於 8週齡 (致病性) or 屠宰前21天雞隻
- 產蛋中避免使用



- 生物安全
  - 清潔、消毒 避免其他免疫抑制病原
  - 預防二次性感染

家禽重要疾病診斷研判圓桌會議結論

雞傳染性貧血現況與對策：

- 落實生物安全措施。雖然一般消毒劑消滅 CAV 的效果有限，但可降低協同或繼發感染病原的誘因，故仍具價值性。
- 種雞場須進行定期之CIA 血清抗體力價監測。
- 對於曾有 CIA 病史之污染種雞場，建議其免疫工作宜在 8-10 週齡時實施，力價的維持則可利用野外毒做自然補強。對於 CIA 壓力較輕場之免疫工作則可接近開產前 4-6 週進行，以利母雞於產蛋期維持高的 CIA 血清抗體力價。

- 加強孵化場新生小雞之監控，以杜絕 CIA 垂直傳染情形，確保雛雞品質。
- 目前市售活毒疫苗雖為減毒疫苗，依仿單說明不宜在 8 週齡前使用，本病仍以種禽免疫藉由移行抗體保護雛禽為主。故種禽業者應落實雞傳染性貧血之免疫計畫，提高種禽血清抗體力價之均勻度，以降低垂直傳染發生。操作疫苗時，應注意使用時程及生物安全措施。
- 建議於種雞 8-16 週齡時接種疫苗，並於免疫後 3-4 週檢測抗體力價，若抗體力價不足或均勻度不佳時，最遲應於開產前 4 週補強。
- 雞傳染性貧血常與球蟲症或其他病原混合感染，故養禽業者在防治本疾病時，應加強監控其他疫病，並應積極防治免疫抑制性疾病，避免影響疫苗接種的效果。